Safe System of Work

Use of 360° Excavators

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O v e r v i e w

The following information provides guidance on the company's Standard Operation Procedures for works involving the use of 360° Excavators on site.

All employees have a responsibility for their safety and the safety of others by adhering to these codes of practice.

The Operator must always refer to the task Risk Assessment prior to starting works.

Hazards Identified

- Adverse weather
- Crushing
- Machine Rollover
- Cuts and abrasions
- Vehicle movements
- Falls from height
- Manual handling
- Slips & Trips

Personal Protective Equipment (PPE) Requirements

Generally there are two types of 360° hydraulic excavators, one being mounted on tracks and the other mounted on wheels (rubber duck). You will also find rope operated excavators (Draglines) which are generally mounted on tracks. It must always be ensured that when the machine is being operated in confined places or near other site personnel there is clearance of at least 600mm for tail swing, the danger area must be barricaded if necessary.

If it is necessary for anyone to go into the machines working radius whilst it is working then the machine operator must he made aware by signals or other means such as radio before doing so. No person should be permitted at any time to be within the machines working radius without obtaining permission from you the operator.

No excavator bucket or load should he slewed directly over personnel, work huts or vehicle cabins. Vehicles should be loaded over the side or rear and the material should not be dropped from an unnecessary height.

The manufacturer's recommended bucket size must not be exceeded. It may be possible to lift the load with the bucket in close to the machine at ground level but as the load radius and elevation Increases, the lifting capacity of the machine decreases.

Dangerous overhangs must not be created on a high work surface, the wheel, or tracks of the machine should be positioned at 90° to the workface to enable quick withdrawal if necessary. In excavating the type of soil should be taken into account in determining a safe position for the machine.

If operating on a gradient cannot be avoided, it must be ensured that the working cycle is slowed down, that the bucket is not extended too far in the downhill direction and that travel is undertaken with extreme caution.

A large excavator must never be permitted to travel in a confined area, or around people without a banks person to guide the driver, who should have the excavator attachment close to the machine with the bucket just clear of the ground.

On wheeled excavators it is essential that the tyres are in good condition and correctly inflated. If stabilising devices are fitted they should be used when the machine is excavating.

Excavators should only be driven by fully trained, competent operators who have been authorised to do so and any operation should be accompanied by a suitable risk assessment. The nationally recognised qualification for competence within the UK is an NVQ Level 2 in plant operations.

For lifting operations, operators must also provide proof of both training and competence for using excavators as cranes. This will include knowledge and skills relating to: pre-lift checks; differences between various lifting accessories or machine attachments (e.g. quick hitches); load connection procedures; interpreting the machine's lifting duties limitations at varying radii, over the front, side and rear of the machine travelling with a load; and load disconnection procedures.

Pedestrians must be segregated from the 'operational area' of the excavator and its transport routes; these routes should have good clear signage and appropriate traffic control measures. The use of hands-free mobile technology should be considered to maintain good communication at a safe distance with site management and/or slingers during any lift or carry operations.

Site managers should be knowledgeable with regards to safe lifting operations and must take an active role in planning, monitoring, controlling and managing all lifting operations on site. The lift plan and method statement should be communicated to all operatives involved.

Vehicle routes should avoid significant obstacles, uneven ground and inclines, particularly if across the direction of travel. Where the route is near to a drop, such as an excavation or embankment, extra support or barriers should be provided for the edge of the route.

Before using an excavator to lift (see SSoW-020), a thorough risk assessment should be undertaken to determine whether alternative work processes could be used to avoid lifting operations or whether a different item of equipment would represent a safer choice.

The operator has a duty to conduct maintenance and inspection at the start of each working shift (for example, of chains, shackles, hooks and slings) to ensure they are suitable for the lift, clearly marked with a safe working load and in good working condition. A full and complete record of all maintenance and inspection undertaken must be kept and any defective machinery or lifting gear found should not be used under any circumstance.

The excavator must always be operated within its capability and if in any doubt, the operator should consult with the operations and maintenance manual supplied by the Original Equipment Manufacturer

(OEM).

Excavators that are used for lifting are subject to a thorough examination and test at least every 12 months. Machines must also be fitted with an audio-visual safe load indicator and lift capacity at various radii should be clearly stated. Lifting eyes should be closed, not open and rated by the OEM.

When travelling, machine stability will be improved by reducing travel speeds, crowding the load into the machine as far as is safe/practicable and keeping the load low to the ground. When placing the load into position, engine revs should be lowered to tortoise mode and the machine’s boom and dipper manipulated to move the load into position. At all times, the load must remain freely suspended.